ENGINEERING
McMaster-Mohawk
Bachelor of Technology
Partnership

McMaster University

Course Outline

1. COURSE INFORMATION

Session Offered	Fall 2026	
Course Name	Artificial Intelligence and Machine Learning	
Course Code	SMRTTECH 4AI3	
Date(s) and Time(s) of lectures	We 12:30PM - 3:20PM	
Program Name	Automation Engineering Technology	
Calendar Description	This course deals with concepts of Artificial Intelligence and Machine Learning. It covers supervised classification based on artificial neural networks - deep Learning, as well as unsupervised learning - clustering, regression, optimization and generative learning. In addition, the course covers application in artificial intelligent systems.	
Instructor(s)	Dr. Zhen Gao	E-Mail: gaozhen@mcmaster.ca Office Hours & Location: ETB-516 Thursday 1:00 PM – 2:30 PM or upon request
	Mr. Siqi Zhao	E-Mail: zhaos98@mcmaster.ca

2. COURSE SPECIFICS

Course Description

This course provides both breadth and depth in terms of Artificial Intelligence and Machine Learning with a good mixture of theory, practice and real world application. This course covers the fundamentals and hands-on modules of classic and advanced AI and ML techniques. The students will understand various AI and ML models and tools such as gradient decent, global searching, SVM, linear regression, logistic regression, shallow neural networks, DNN and its variants. The skill of management of training data will also be taught. Python and its eco-system including Numpy, SciPy Matplotlib, scikit-learn, OpenCV-Python, TensorFlow, Keras will be learnt/applied. Students should know how to program classic AI/ML algorithms from scratch without just relying on importing the 'plug-and-play' libraries.

Instruction Type

Code	Type	Hours per term
C	Classroom instruction	39
L	Laboratory, workshop or fieldwork	39
T	Tutorial	
DE	Distance education	
	Total Hours	78

Resources

ISBN	Textbook Title & Edition	Author & Publisher
ISBN: 9781108470049	Mathematics for Machine Learning	*M. Peter Deisenroth, A. Faisal, and C. Soon Ong*
ISBN: 0387310738	Pattern Recognition and Machine Learning	*Christopher Bishop*
Other Supplies	**Source**	
Lecture Notes		

Prerequisite(s)	*ENGTECH 2MT3, 1CP3, 1PR3, 4EE0, SMRTTECH 4ID3 and registration in level IV of Automation Systems Engineering Technology, OR MECHENG 4R03 or IBEHS 4A03 and registration in any Mechanical Engineering Program*
Corequisite(s)	
Antirequisite(s)	*SFWRTECH 4AI3, MANTECH 4AI3*
Course Specific Policies	All work must be shown to get full credit. Missed test marks due to legitimate reasons (officially verified by the faculty of Engineering/B.Tech) will be added to the final examination. Labs will be conducted in-person. Completed labs should be uploaded to the lab's drop box before midnight of the due date. Labs missed due to legitimate reasons must be completed later mutually agreed with the instructor.
Departmental Policies	Students must maintain a GPA of 3.5/12 to continue in the program. In order to achieve the required learning objectives, on average, B.Tech. students can expect to do at least 3 hours of "out-of-class" work for every scheduled hour in class. "Out-of-class" work includes reading, research, assignments and preparation for tests and examinations. Where group work is indicated in the course outline, such collaborative work is mandatory. The use of cell phones, iPods, laptops and other personal electronic devices are prohibited from the classroom during the class time, unless the instructor makes an explicit exception. Announcements made in class or placed on Avenue are considered to have been communicated to all students including those individuals that are not in class. Instructor has the right to submit work to software to identify plagiarism.

3. SUB TOPIC(S)		
Week 1	Introduction of AI and ML	
Week 2	Python ecosystem, data handling	
Week 3	Global searching	
Week 4	Linear regression and polynomial regression	
Week 5	Logistic regression	
Week 6	K-nearest neighbors classifier and K-means clustering	
Week 7	Support vector machine	
Week 8	Shallow neural networks	
Week 9	Deep learning	
Week 10	Generative modeling, GANs	
Week 11	Application of AI and ML	
Week 12	Project presentation	
Week 13	Final review	
Midterm Recess: Monday, October 12 to Sunday, October 18 Classes end: Thursday, December 10 Final examination period: Friday, December 11 to Wednesday, December 23		
List of experiments		
Lab 1	Python Basics	

Lab 2	Linear Regression
Lab 3	Data Pre-processing & Classification
Lab 4	Feature Selection & Regression
Lab 5	Multi-class classification & Cross-validation
Lab 6	K-means & Clustering
Lab 7	Neural Networks I
Lab 8	Neural Networks II
Lab 9	Convolutional Neural Networks
Lab 10	Generative Adversarial Network (GAN)
	Lab Test

Note that this structure represents a plan and is subject to adjustment term by term.
The instructor and the University reserve the right to modify elements of the course during the term. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes.

4. ASSESSMENT OF LEARNING *including dates*	**Weight**
Assignments	20%
Labs	20%
Lab test	15%
Project	20%
Final examination (tests cumulative knowledge)	25%
TOTAL	**100%**

Note that the percentage distribution of each module may be adjusted.
Percentage grades will be converted to letter grades and grade points per the University calendar.

5. LEARNING OUTCOMES

1. Be familiar with classic AI and ML methods as well as newly emerged techniques such as Deep Learning.
2. Be capable of connecting theory with practice through transferable skills.
3. Understand the current limitations of AI and ML when applying in real world.
4. Know how to properly and efficiently handle AI/ML data.
5. Gain experience to design, evaluate and improve AI/ML models for real-world problem solving such as object classification/recognition, system prediction especially in area of automation.
6. Be familiar with well-known methods such as gradient decent, global searching, SVM, linear regression, logistic regression, shallow neural networks, DNN and its variants.
7. Be able to code and debug independently of classic AI/ML methods from scratching without using libraries or tools.
8. Be able to apply advanced libraries or tools, such as Tensorflow, Keras, scikit-learn, OpenCV-Python, for complicated AI/ML problems.

6. COURSE OUTLINE – APPROVED ADVISORY STATEMENTS

ANTI-DISCRIMINATION

The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as soon as possible.

http://www.mcmaster.ca/policy/General/HR/Discrimination_Harassment_Sexual_Harassment-Prevention&Response.pdf

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/

The following illustrates only three forms of academic dishonesty: The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

COMMUNICATIONS

It is the student's responsibility to:

- Maintain current contact information with the University, including address, phone numbers, and emergency contact information.

- Use the University provided e-mail address or maintain a valid forwarding e-mail address.
- Regularly check the official University communications channels. Official University communications are considered received if sent by postal mail, by fax, or by e-mail to the student's designated primary e-mail account via their @mcmaster.ca alias.
- Accept that forwarded e-mails may be lost and that e-mail is considered received if sent via the student's @mcmaster.ca alias.
- Check the McMaster/Avenue email and course websites on a regular basis during the term.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests. http://www.mcmaster.ca/policy/Students-AcademicStudies/Studentcode.pdf

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.